

Present document is related to erytra eflexis Service Manual (REGD-0019974)

Instrument:	erytra eflexis	Form version:	7.0
Serial number:		Code:	
Installation date:			

Customer Site:	Customer Name:
Department:	
Tel:	E-Mail:

Service Engineer Name:
Department:
Tel:
E-Mail:

Action		Checked / Done	Value	Comment
Instrument decontaminated by the User / Operator as described in Instructions for Use decontamination chapter.				
INITIAL CHECKS				
Visual inspection of (every year):				
General appearance				
Output cables and connectors				
Grounding continuity test:	Screw attaching centrifuge 1 to chassis			
Switch off and unplug the electrical appliance. Check with a multimeter electric continuity between earth terminal of IEC inlet appliance and:	Screw attaching centrifuge 2 to chassis			
SOFTWARE				
Inspection of the following items (every year):				
Folders inspection				
COMPUTER AND TOUCHSCREEN				
Inspection of the following parts (every year):				
SSD available space				
Touchscreen operability (touch functionality in all screen areas)				

Action	Checked / Done	Value	Comment
PARTS REPLACEMENT ⁽¹⁾			
Number of tests of last year PM:			
Current number of tests:			
Every 80.000 tests ⁽⁵⁾ ⁽⁴⁾			
Probe (PR) ⁽⁶⁾			
Every 160.000 tests ⁽⁵⁾ ⁽²⁾			
MK Syringe (SYR) and MK valve			
CAT Z axis chain			
Probe washer (WS) ⁽⁷⁾			
Every 240.000 tests ⁽⁵⁾ ⁽³⁾			
Toothed belt T2.5/245			
o 2 units in PIP arm			
o 2 units in CAT arm			
Aspiration pump VPB			
Every 400.000 tests ⁽⁵⁾			
SOL B dispensation pump (PB)			
Every 3 years (independently of the number of processed tests)			
Dilution cup (DC)			
Teflon tube 2-way adaptor (F70)			

(1) See Annex 1 for parts replacement procedures

(2) **Note:** In case previous P/N 234132 TREADER MOTOR is still installed, it should be replaced with the current model.

(3) **Note:** In case previous P/N 234187 DISPENSING PUMP is still installed, it should be replaced with the current model.

(4) **Note:** In case previous P/N 233952 HAMILTON SYRINGE 1ml and 233953 HAMILTON VALVE are still installed, it should be replaced with the current model.

(5) See Annex 4 table equivalence number of tests/year under instrument standard use.

(6) **Note:** 80.000 tests or 3 years maximum

(7) **Note:** 160.000 tests or 3 years maximum

Action	Checked / Done	Value	Comment
GENERAL MODULE			
MONITOR ARM			
Mechanics (every year):			
Check that the arm movement is smooth			
Check that the arm blocking screw prevents undesired impacts of the monitor against the instrument chassis			
DOOR			
Mechanics (every year):			
Check that movement is smooth			
Check that door does not hit during automatic opening			
Check that the latch enters smoothly when closing			
Lubricate door spring and door guides (grease) ⁽⁸⁾			
OTHER			
If UPS is connected, check battery level			

⁽⁸⁾ See Annex 2 for instructions on all lubrication actions.

Action	Checked / Done	Value	Comment
CARDS MODULE GATE			
Mechanics (every year):			
Check that movement is smooth			
SAMPLES AND REAGENTS ENTRY (SRE)			
Mechanics (every year):			
Perform a general cleaning			
Check sample tube holders			
Check reagent vial/bottle lodgings			
CARD DRAWERS			
Mechanics (every year):			
Perform a general cleaning			
Check card racks centerers and retainers			
Lubricate drawers latch (grease) ⁽⁸⁾			
Lubricate linear guides (grease) ⁽⁸⁾			
DILUTION CUP			
Mechanics (every year):			
Perform a general cleaning			
Check dilution cup drain tubing			
Electronics/functional (every year):			
Check proper stirrer operation (PROFILE_0: 400 ± 40 rpm)			

Action		Checked / Done	Value	Comment
INCUBATOR MODULE				
Mechanics (every year):				
Perform a general cleaning				
CENTRIFUGE MODULE				
Mechanics (every year):				
Clean pot of the centrifuge ⁽⁹⁾				
Check rotor and card holders status ⁽⁹⁾				
Check that rotor movement is smooth ⁽⁹⁾				
Centrifugation verification (PROFILE_0)	Speed: 1206 ± 36 rpm)		CEN 1:	
	Time: 550 ± 10 s (including braking time)		CEN 2:	
			CEN 1:	
			CEN 2:	

⁽⁹⁾ See Annex 5 for instructions to remove centrifuge pot.

PIP MODULE				
Mechanics (every year):				
Perform a general cleaning				
Check belts tension and status				
Check chains, wiring and tubing status				
Lubricate X-axis linear guide (oil) ⁽⁸⁾				
Lubricate Y-axis linear guide (oil) ⁽⁸⁾				
Lubricate horns linear guide (oil) ⁽⁸⁾				
Lubricate probe leadscrew (grease) ⁽⁸⁾				
Clean/check barcode reader mirrors				
Lubricate probe washer holder cam (grease) ⁽⁸⁾				
Lubricate horns rack and pinion system (grease) ⁽⁸⁾				
Electronics/functional (every year):				
Check CPD system				
Check probe offset in TSS				
Run Leak-tightness test in TSS				
CAT MODULE				
Mechanics (every year):				
Perform a general cleaning				
Check belts tension and status				
Check chains and wiring status				
Check gripper pins, centerer and fingers movement/status				

Action	Checked / Done	Value	Comment
CAT MODULE			
Lubricate X-axis linear guide (oil) ⁽⁸⁾			
Lubricate Y-axis linear guide (oil) ⁽⁸⁾			
Lubricate gripper linear guide (oil) ⁽⁸⁾			
Lubricate gripper leadscrew (grease) ⁽⁸⁾			
Clean/check barcode reader mirror			
Clean/check presence detector ⁽¹⁰⁾			
Lubricate gripper pins (oil) ⁽⁸⁾			
Lubricate gripper centering system ⁽⁸⁾			

⁽¹⁰⁾ See Annex 3 for instructions.

Action	Checked / Done	Value	Comment
FLUIDICS MODULE			
Bottle entry positions			
Mechanics (every year):			
Perform a general cleaning			
Check status of the fluidic input/output system for each bottle position			
Check status of the bottles			
Lubricate bottle pin entry for each bottle position (grease) ⁽⁸⁾			
Cards waste tray area			
Mechanics (every year):			
Perform a general cleaning			
Check status of the cards disposal system (internal/external).			
Fluidic circuit			
Mechanics (every year):			
Perform a general cleaning			
Check tubing			
Check vacuum filters status			
Check that the main fluidic manifold has no scratches nor leakages			
Check that interconnection manifold has no scratches nor leakages			
READER MODULE			
Mechanics (every year):			
Check/clean reader pattern ⁽¹¹⁾			
Check/clean reader camera optics ⁽¹¹⁾			
Check reader foam status ⁽¹¹⁾			

⁽¹¹⁾ See Annex 5 for instructions to remove reader cover.

Replaced parts				
Spare part number	Description	Lot number	Quantity	Comments

Date: _____

Service Engineer signature: _____ Customer signature: _____